

3	(a)	Copper has one conduction electron per atom. The density of copper is 8960 kg m^{-3} . The mass of one mole of copper is 63.5 g. Show that the number density of charge carriers in copper is $8.49 \times 10^{28} \text{ m}^{-3}$.	
			[3]
	L2	Volume of per mole of copper = mass per mole / density = $(63.5 \times 10^{-3}) / 8960$ = $7.0871 \times 10^{-6} \text{ m}^3$ Number of conduction (mobile) electrons per mole of copper = 1 electron per atom x number of atoms per mole = 1 electron per atom x Avogadro's constant = $1 \times (6.02 \times 10^{23}) = 6.02 \times 10^{23}$ Number density of charge carriers = $6.02 \times 10^{23} / 7.0871 \times 10^{-6}$ = $8.49 \times 10^{28} \text{ m}^{-3}$ (shown)	M1 M1 M1 A0

- (b) A composite wire XYZ is made by connecting in series two uniform wires, each of length L and made of copper but having different diameters as shown in Fig. 3.1. One wire has diameter d and the other wire has diameter $2d$.

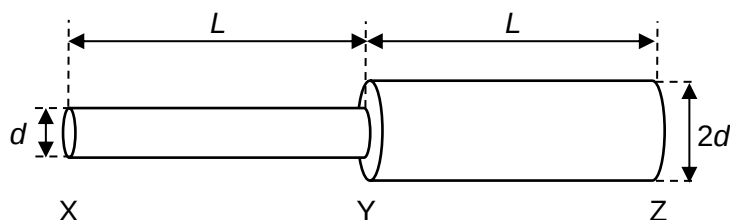


Fig. 3.1

A potential difference is then applied across X and Z of the wire and a current flows through the wire.

On Fig. 3.2, sketch a graph to show how the drift velocity v_d of electrons through the composite wire varies with distance along the wire from end X to end Z.

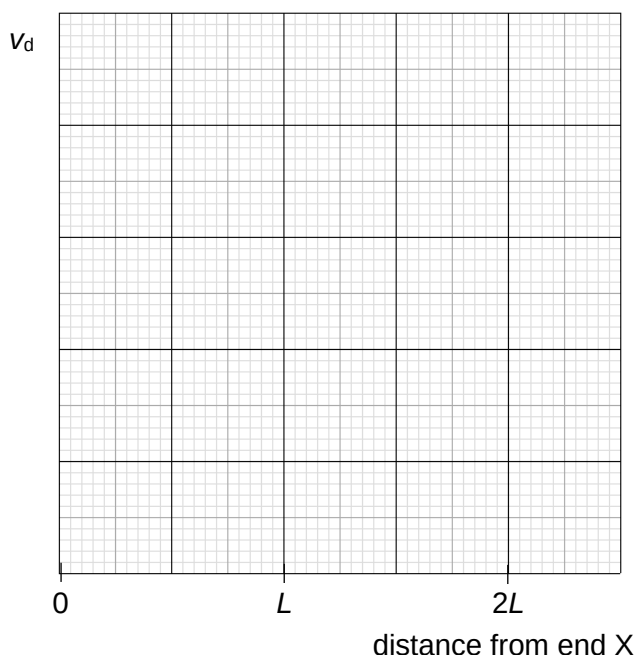


Fig. 3.2

[3]

L2

$$I = neAv_d$$

$$v_d = \frac{I}{neA} = \frac{I}{ne\left(\frac{\pi D^2}{4}\right)} = \frac{4I}{\pi neD^2}$$

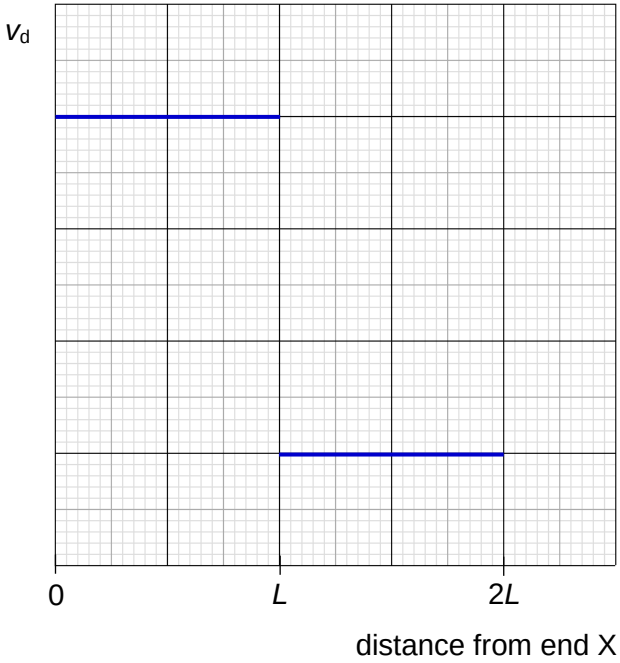
Since current I is constant along the two wires connected in series, and, since n and e are the same for the same material,

$$v_d \propto \frac{1}{D^2}$$

Therefore,

in the wire where diameter D is twice as large, the drift velocity v_d is one-quartered.

In the same wire where diameter is constant, v_d is constant.

		 Marks scheme: v_d is <u>constant</u> in each wire. v_d in the wire XY is <u>4 times greater</u> than v_d in wire YZ. - Graph only drawn in the range of distance 0 to $2L$, and, graph should be big enough, i.e. span at least half the graph grid in both horizontal and vertical directions.	B1 B1 B1
	(c)	The mean speed of a conduction electron in the wire is very much greater than the drift velocity of the conduction electrons in the wire. Explain this observation. 	
	L3	The conduction electrons <u>experience electric forces/accelerations in all directions as a result of collisions with the lattice ions</u>. Thus the individual conduction electrons have a <u>range of velocities</u> of different magnitudes as well as directions, and the electron paths are rather random/erratic. The <u>mean speed, regardless of direction</u>, is large. Whereas drift velocity is a <u>vector quantity</u> and an <u>average velocity</u>, which is quite small since there is a <u>large number</u> of conduction electrons and their individual velocities are rather <u>random</u> in direction and magnitude.	[2] B1 B1

[Total: 8]

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| 4 | A mass m is suspended from a vertical spring of spring constant k attached to a fixed support |
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